

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Volume of solid = 8 cm^3 (1) Substitution: $\frac{26}{8}$ (1) [Allow substitution mark with 40 or 48 cm^3] = $3.25 \text{ [g/cm}^3\text{]}$ (1) Use of 40 cm^3 ($\rightarrow 0.65 \text{ g/cm}^3$) or 48 cm^3 ($\rightarrow 0.54 \text{ g/cm}^3$) : 1 mark NB. Allow ecf on incorrect volume of solid only if calculation of volume shown, e.g. $48 - 40 = 4$ leading to $6.5 \text{ [g / cm}^3\text{]}$	1	1 1		3	3	3
	(b)			Method of submerging object [e.g. sinker or pencil point] (1) Determination of <u>total</u> volume displaced by object and submerging object (1) Accounting for the volume of the submerging object [e.g. subtracting volume of sinker / <u>just</u> submerging by pencil point](1) Use of liquid of lower density in which the object sinks $\rightarrow$ first mark only.			3	3		3
	(c)			Particles in solids are <u>more</u> closely packed [or most solid {molecules / particles} are heavier than water molecules] (1) so more {contained / mass} per unit volume (1)	2			2		
				Question 3 total	3	2	3	8	3	6